PRODUCT USER MANUAL

OMI_SI
ANTARCTIC_OMI_SI_extent_obs

Issue 2.2

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Approval Date: June June 20202023

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Change Record

Issue	Date	§	Description of Change	Author	Validated By
1.0	25.01.2018	all	Creation of the document	Thomas Lavergne	Cecilie Wettre
2.0	12.04.2019	All	Revision for extension to 2018, and using OSISAF products OSI-450 and OSI-430-b as source.	Thomas Lavergne	Cecilie Wettre
2.1	28.04.2020	All	Revision for extension to 2019	T. Lavergne	C. Wettre
2.2	20.06.2023	All	Revision for 2023 release, use OSI SAF OSI-420 v2.2 as source	T. Lavergne	C. Wettre

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GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
MOI	Mercator Océan International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
TAC	Thematic Assembly Centre
SI	Sea Ice
CDR	Climate Data Record
CF	Climate Forecast (convention for NetCDF)
ICDR	Interim Climate Data Record
NRT	Near-Real-Time
OSISAF	EUMETSAT's Ocean and Sea Ice Satellite Application Facility
RMS	Root mean square
SIC	Sea Ice Concentration
SIE	Sea Ice Extent
PC	Production Center
PU	Production Unit
ftp	Protocol to download files
OpenDAP	Open-Source Project for a Network Data Access Protocol. Protocol to download subset of data from a n-dimensional gridded dataset (ie: 4 dimensions: lon-lat,depth,time)
Subsetter	CMEMS service tool to download a NetCDF file of a selected geographical box using values of longitude an latitude, and time range

I INTRODUCTION

I.1. Summary

This document contains the Product User Manual for the Copernicus Marine Ocean Monitoring Indicator (OMI) ANTARCTIC_OMI_SI_extent_obs. The data are monthly averaged Sea Ice Extent (SIE) time series.
The OMI product is a reformatting of the Southern Hemisphere monthly SIEs from the Sea Ice Index (OSI-420) product from the EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSI SAF), developed and delivered by MET Norway and the Danish Meteorological Institute (DMI).
The EUMETSAT OSI SAF acknowledges the R&D contribution from the ESA Climate Change Initiative projects.

I.1.1. The OSI SAF OSI-420 Sea Ice Index

The EUMETSAT OSI SAF Sea Ice Index product (OSI-420) used in the CMEMS OMI is fully described in the corresponding OSI SAF document:

- Product User Manual for the OSI SAF Sea Ice Index OSI-420, version 2.2, 31/05/2023 (accessible via https://osisaf-hl.met.no/v2p2-sea-ice-index)

In June 2023, OSI-420 was updated to v2.2. The OSI-420 v2.2 product is computed from three SIC products:

- The "v3" Climate Data Record (CDR), covering Oct 1978 to Dec 2020 (OSI-450-a)
- The "v3" Interim Climate Data Record (ICDR), starting in Jan 2021, with a 16 days latency (OSI-430-a);
- A Near-Real-Time (NRT) product that uses the same processing chain as OSI-450-a and OSI-430-a, for the most recent 15 days (up-until yesterday).

Note: the previous version of this CMEMS OMI was based on the v2 SIC products : OSI-450 and OSI-430-b.

I.1.2. The Copernicus Marine OMI Sea Ice Extent

The CMEMS OMI Sea Ice Extent data file is a reformatting from the OSI-420 product file, in particular adding pre-computed climatological monthly SIE. Also, the OMI has a yearly update frequency while OSI-420 has daily or monthly update frequency.
In addition, a high-quality figure and a scientific context document are prepared specifically for the CMEMS OMI.
Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.
Product OMI_SI_extent is described in the Quality Information Document (QUID) CMEMS-OMI-QUID-SIE.pdf (http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-SIE.pdf).

I.2. History of change

12.04.2019	• Publication of ANTARCTIC_OMI_SI_extent_obs product • Use new version of OSI SAF SIC as input.
28.04.2020	Extension to 2019
20.06.2023	Extension through 2022, using OSI SAF OSI-420 v2.2 as input.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1. General Information about products

The time-series starts in January 1979 and ends in December the year previous to the release date (e.g. release 2023 holds data up until December 2022).

Product(s) name	ANTARCTIC_OMI_SI_extent_obs		
Geographical coverage	Ice-covered oceans of the Southern Hemisphere. Lake ice is not included.		
Variables	Monthly mean sea ice extent, and its climatology by month.		
Update frequency	Yearly (released Q3 YYYY)		
Available time series	January 1979 – December (YYYY-1)		
Delivery mechanism	FTP		
Temporal/Spatial resolution	Monthly/aggregated		
Integration depth	Surface		
Format	NetCDF CF1.7		

II.2. Climatology period

For the ANTARCTIC OMI a climatology by month is also provided. The climatology period is the whole period.

II.3. Details of the datasets

For each product, one table with details of datasets content

II.3.1. ANTARCTIC_OMI_SI_extent_obs

ANTARCTIC_OMI_SI_extent_obs		
artic_omi_si_extent_obs		
	contains the 1D monthly Southern Hemisphere Sea Ice Extent time series, and its climatology monthly	
	siextent [km2] Monthly averaged Sea Ice Extent sea_ice_extent	
	monthlyclimsiextent_mean [km2] Climatology (by month) of the Sea Ice Extent (mean value) sea_ice_extent	
	monthlyclimsiextent_std [km2] climatology (by month) of the Sea Ice Extent (one standard deviation) sea_ice_extent	

II.4. Product System Description

For the Antarctic – An indicator that gives the monthly mean sea ice extent (SIE in km²) for the southern hemisphere. The time-series starts in January 1979 and ends in December the year previous to the release date (e.g. release 2023 holds data up until December 2022).

III Download a product

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.
Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

IV FILES NOMENCLATURE and FORMAT

IV.1. Nomenclature of files

OMI_SI_extent files nomenclature when downloaded through the CMEMS FTP Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>_R<climatologyrange>

Where :

- **<datasetname>** is the dataset reference.
- **<timestep>** is the first time step contained in the file.
- **<productiontime>** is the production time when the file was generated.
- **<climatologyrange>** is the range of years used for the climatology
- **.nc** is the standard NetCDF filename extension.

IV.2. File Format

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

IV.3. Reading Softwares

The products are stored using the NetCDF-CF format. Note that these data can be browsed and used through several softwares, like:

- ncBrowse: http://www.epic.noaa.gov/java/ncBrowse/
- NetCDF Operator (NCO): http://nco.sourceforge.net/
- http://www.unidata.ucar.edu/software/netcdf/

V References

See the Quality Information Document (QUID) CMEMS-OMI-QUID-SIE.pdf (https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-SI-extent-obs.pdf).

OSI SAF OSI-420 v2.2 webpage and documentation https://osisaf-hl.met.no/v2p2-sea-ice-index